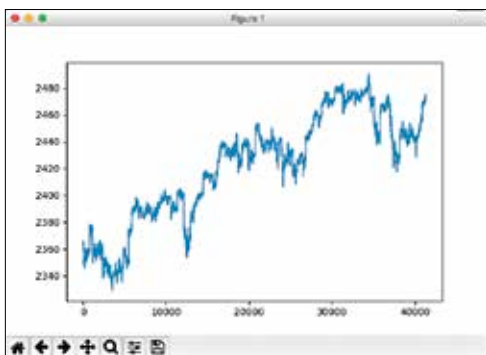




Stock price prediction using deep learning.

of problems. Early adopters have demonstrated success in business outcomes like fraud detection, manufacturing performance optimisation, preventative maintenance, and recommendation engines. Some business applications of deep learning include text summarisation,

text automation or next text suggestion, language translation, call center analysis, enterprise application development and digital asset management in marketing.

Additionally, businesses also gain multiple positive outcomes and automate multiple time consuming processes through deep learning with applications such as cybersecurity defense, recruiting automation, smart conversational interfaces, reduced energy use and costs, predicting vulnerability of a piece of software, becoming more customer-centric, market prediction, accounting and more.

Some popular CNNs used in deep learning

Convolutional Neural Network (CNN) is a unique, multi-layer neural network which is specially created to recognise visual patterns straight from pixel-form images with negligible preprocessing. CNNs will be covered in a more comprehensive manner in Chapter 04. They are similar to Neural Networks, but the difference is that CNNs' architecture are specifically designed for images as input. The layers of a CNN have their neurons arranged in three dimensions - width, height and depth. They are exceptionally useful for spatial data, object identification and image analysis since it uses multidimensional neuron structures.

Deep learning itself owes its increasing popularity lately to CNNs since it powers notable deep learning projects like self-driving cars, drones, text analytics and computer vision. Here's a list of some of the most popular CNNs used in conjunction with deep learning and its most illustrious applications.